

Relative-Periodic Coupled Graphs

1 Definitions

Let

$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}).$$

Let smooth periodic functions u, v solve

$$u_t = \partial_x \left(\frac{u_x}{\sqrt{1+u_x^2}} \right) - (u-v), \quad v_t = \partial_x \left(\frac{v_x}{\sqrt{1+v_x^2}} \right) + (u-v). \quad (1)$$

Suppose that, for some

$$T > 0, \quad a \in \mathbb{T}, \quad b \in \mathbb{R}, \quad (2)$$

one has

$$u(x, T) = u(x + a, 0) + b, \quad v(x, T) = v(x + a, 0) + b. \quad (3)$$

2 Best Known Result

Horizontal translations and common vertical translations are symmetries of the functional

$$\mathcal{L}(u, v) = \int_{\mathbb{T}} \left(\sqrt{1+u_x^2} + \sqrt{1+v_x^2} + \frac{1}{2}(u-v)^2 \right) dx. \quad (4)$$

The source indicates that an exact Lyapunov identity, together with rigidity of the critical points, should determine the classification. It does not provide the resulting classification.

3 Problem Formalization

Classify all such smooth solutions and all possible triples (a, b, T) .